



Data-Driven Prediction of SAnMBR Performance under Phenolic Shock Loads: A Comparative Machine Learning Study

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Abstract

Treating phenolic wastewater remains a major challenge due to the toxicity and inhibitory effects of compounds such as 2,4-dichlorophenol (2,4-DCP), which destabilize biological processes. This study investigates the performance of a pilot-scale submerged anaerobic membrane bioreactor (SAnMBR) treating 2,4-DCP-contaminated wastewater over 189 days and develops a data-driven machine learning framework to predict COD removal efficiency. Based on 189 experimental observations and six key operational parameters, three algorithms, namely multiple linear regression, artificial neural networks, and support vector regression were trained. SVR outperformed the other models, achieving a test R^2 of 0.952, an RMSE of 2.96, and strong generalization performance. The results indicate that optimal COD removal occurred within a narrow pH range (8–9) and at low 2,4-DCP loads (<50 mg/L), with steep performance declines observed under inhibitory concentrations (>150 mg/L). In addition to COD, SAnMBRs also achieved high removal efficiencies for turbidity (>75%), TSS (>90%), and 2,4-DCP (>80%). These findings demonstrate that integrating long-term experimental data with interpretable machine learning enables reliable performance prediction and provides actionable insights for optimizing the SAnMBR operation in treating toxic industrial wastewater.

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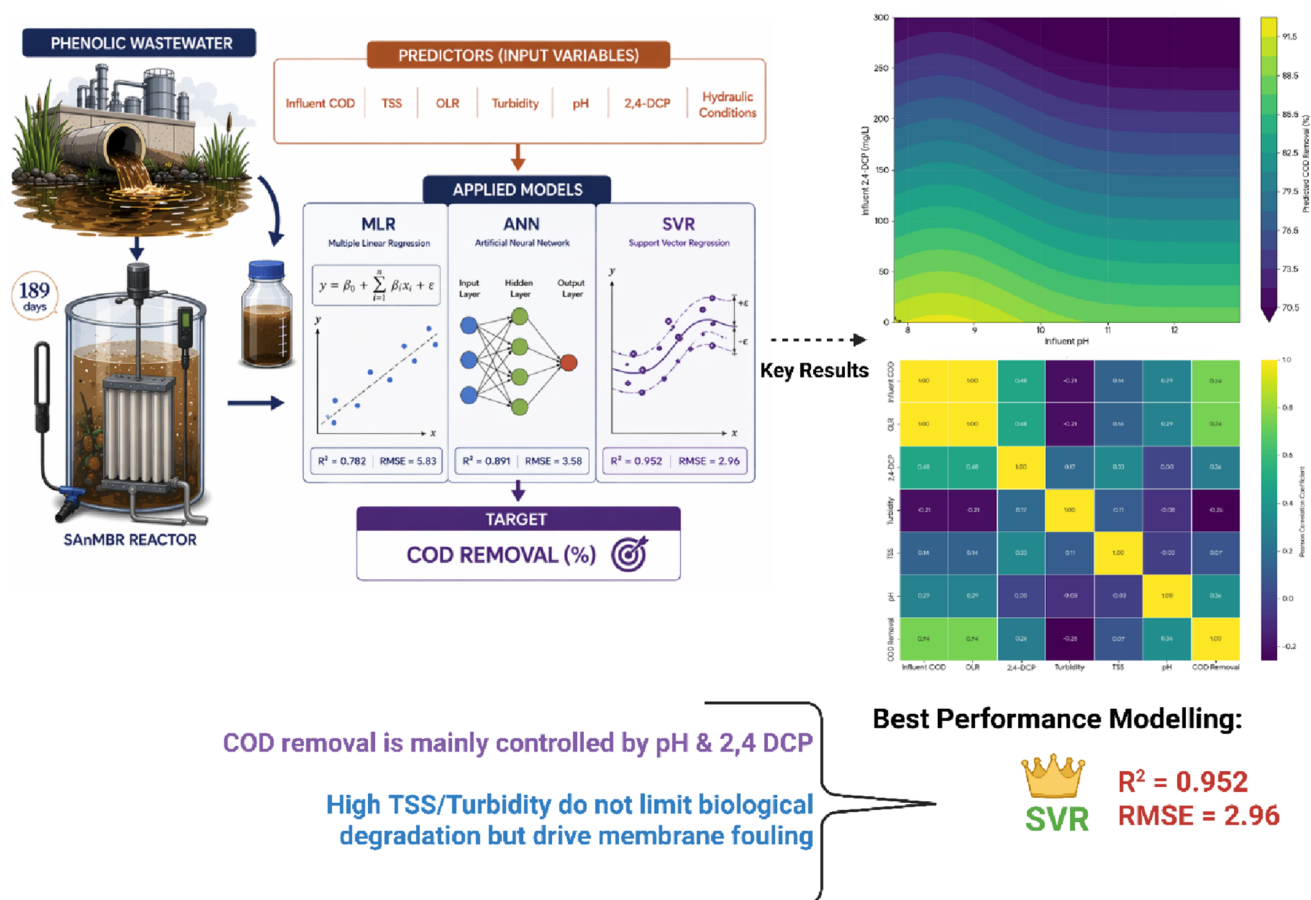
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Graphical Abstract



This graphical abstract presents a structured visual pathway linking long-term experimental operation with data-driven predictive modelling for a submerged anaerobic membrane bioreactor (SANMBR) treating phenolic wastewater. The first segment illustrates the pilot-scale system, highlighting the continuous operation under progressively increasing 2,4-dichlorophenol (2,4-DCP) shock loads over 189 days. Key operational variables, including influent COD, organic loading rate, turbidity, total suspended solids, pH, and 2,4-DCP concentration, are introduced as the primary data inputs. The second segment focuses on the analytical and modelling framework, where experimental data are processed and fed into three machine learning models (MLR, ANN, and SVR). The model comparison is visually emphasized to demonstrate the superiority of the SVR model in capturing non-linear system behaviour. The final segment summarizes the key outcomes by linking model predictions with process understanding. It highlights the existence of a narrow optimal operational window governed by pH and 2,4-DCP concentration, where maximum COD removal efficiency is achieved. Additionally, it distinguishes between biological performance drivers and operational constraints, particularly membrane fouling associated with suspended solids. Overall, the graphical abstract provides a concise integration of experimental operation, machine learning modelling, and process interpretation, enabling rapid understanding of how data-driven approaches can enhance the prediction, optimization, and control of SANMBR systems treating toxic industrial wastewater.

Highlight

Long-term pilot SANMBR operated under progressive 2,4-DCP shock loading (189 days).

Stable high removal achieved: COD (80–95%), TSS (>90%), 2,4-DCP (>80%).

SVR outperformed ANN and MLR with high accuracy ($R^2 = 0.952$, RMSE=2.96).

pH (8–9) and 2,4-DCP identified as dominant non-linear drivers.

Optimal performance was confined to low 2,4-DCP (<50 mg/L) and narrow pH window.

Keywords Submerged anaerobic membrane bioreactor · 2,4-dichlorophenol · Phenolic wastewater · COD removal · Support vector regression

1 Introduction

Phenolic compounds, including phenol and its derivatives, represent a class of priority pollutants often found in industrial effluents (e.g., petrochemical, pharmaceutical, dye, and pulp and paper industries). These compounds are characterized by aromatic ring structure bearing one or more hydroxyl groups, which contribute to their chemical stability and solubility, allowing them to persist and disperse in aquatic environment (Catherine et al. 2016). Their persistence, coupled with their bioaccumulative potential, raises severe environmental and public health concerns. Ecotoxicological studies have demonstrated that phenolic compounds can exhibit carcinogenicity, mutagenicity, and endocrine-disrupting effects, even at relatively low environmental concentrations (Menon and Kalyanraman 2025). Quantitatively, phenolic compounds can exert acute toxicity at relatively low concentrations, with reported LC50 values for aquatic organisms typically ranging from 5 to 25 mg/L, while sub-lethal effects such as enzymatic inhibition and oxidative stress have been observed at concentrations below 1 mg/L. Moreover, phenols and chlorophenols are often listed as priority pollutants by regulatory agencies due to their deleterious effects and recalcitrant behaviour. From a human health perspective, phenolic compounds pose a significant risk. For example, chronic exposure through ingestion or dermal contact can lead to neurological, hematological, and hepatic toxicity (Wang et al. 2019). Their removal from wastewater is therefore not only desirable but necessary to avoid environmental accumulation and potential health hazards. However, because of their chemical stability, conventional biological treatment processes sometimes fail to consistently degrade phenols, especially when concentrations are high or the influent composition is complex.

Biological treatment performance is highly dependent on phenolic concentration, as elevated levels can strongly inhibit microbial activity and destabilize reactor performance. In addition, the speciation and toxicity of chlorinated phenolic compounds such as 2,4-dichlorophenol (2,4-DCP) are strongly pH-dependent. Due to its weakly acidic nature ($\text{pK}_a \approx 7.9$), 2,4-DCP exists in both dissociated (ionized) and undissociated (non-ionized) forms depending on the solution pH. At low pH, the non-ionized form predominates, which is more hydrophobic. At lower pH values, the non-ionized form can readily diffuse through microbial cell membranes, thereby increasing its inhibitory effects on biological processes, whereas under neutral to slightly alkaline conditions, the ionized form becomes dominant, reducing membrane permeability and overall toxicity. This pH-dependent behaviour plays a critical role in determining the bioavailability and inhibitory impact of 2,4-DCP in anaerobic systems. Therefore, high-strength phenolic wastewater often requires dilution or

appropriate pretreatment to reduce toxicity prior to biological treatment, ensuring stable and efficient operation (Al-Qodah et al. 2019).

Anaerobic membrane bioreactors (AnMBRs) have emerged as a compelling technology for treating industrial wastewaters, including those containing phenolic compounds. Unlike conventional aerobic treatment, AnMBRs combine anaerobic biodegradation with membrane filtration, enabling high biomass retention, long solids retention time (SRT), and efficient recovery of biogas (Muñoz Sierra et al. 2020). Submerged configurations (SAnMBRs), in which the membrane module is immersed directly in the bioreactor, offer additional benefits such as reduced footprint, lower energy demand, and enhanced process stability. In the context of phenolic wastewater, such systems are particularly attractive because the slow-growing anaerobic microbial communities (e.g., phenol-degrading syntrophic bacteria) can be retained within the reactor over long timescales, maximizing the contact time necessary for degradation (Rezazadeh et al. 2018). These microbial consortia typically include syntrophic bacteria such as *Syntrophus* and *Syntrophorhabdus*, which are involved in the initial degradation of aromatic compounds, as well as methanogenic archaea such as *Methanosaeta* and *Methanosarcina*, which facilitate the final conversion of intermediates to methane (García Rea et al. 2020). Indeed, studies have demonstrated that submerged AnMBRs can achieve high removal efficiencies: in one long-term experiment treating kraft evaporator condensate, chemical oxygen demand (COD) removal exceeded 93–99% with stable methane production, and biogas sparging was effective for controlling fouling (Muñoz Sierra et al. 2020). However, AnMBRs, especially submerged ones, face nontrivial operational challenges. Membrane fouling remains a primary limitation including cake formation, pore blocking, and the accumulation of extracellular polymeric substances (EPS) or soluble microbial products (SMP) can degrade membrane performance, increase transmembrane pressure (TMP), and shorten membrane lifespan (Viet and Jang 2021). In addition, the complex and nonlinear interactions among operational variables, microbial dynamics, and physicochemical properties make predictive modelling and control difficult.

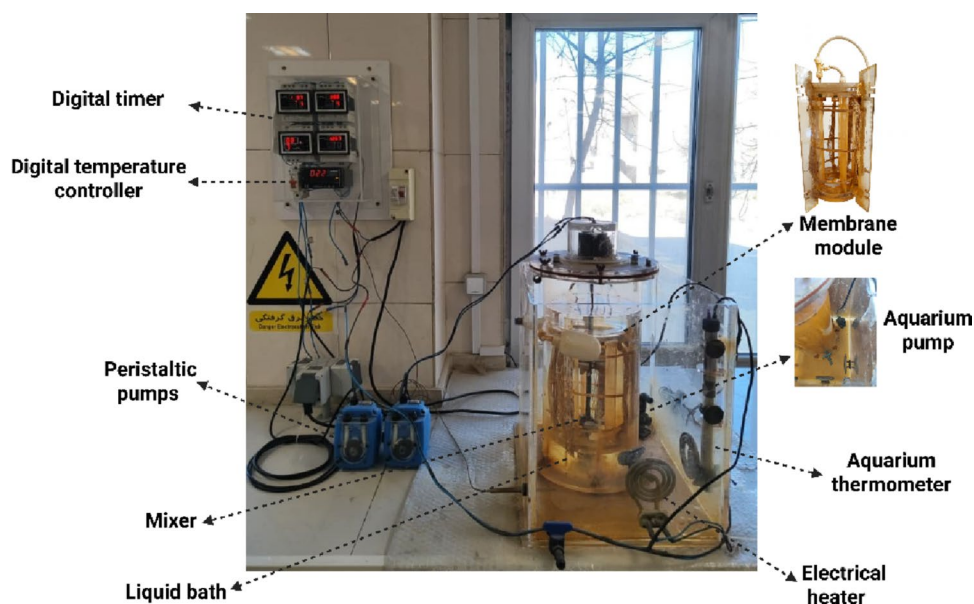
In order to improve efficacy and parameter optimization, machine learning focuses on developing algorithms that learn patterns from data and support prediction or decision-making. In artificial intelligence (AI), machine learning focuses on developing computer algorithms to facilitate intelligent actions through data processing. AI also offers potential for real-time optimization and control, enabling dynamic operational strategies that adapt to changing conditions. However, in SAnMBR systems treating phenolic wastewater, the necessity of ML is not merely

due to its general predictive capability, but arises from the intrinsically complex and non-linear interactions between inhibitory compounds (e.g., 2,4-DCP), microbial activity, and operational parameters such as pH and organic loading rate (OLR), which cannot be adequately captured by conventional mechanistic or linear models. These interactions are further complicated by toxicity thresholds and microbial acclimation effects, where system performance can shift abruptly under phenolic shock loading, making deterministic modelling approaches insufficient. The application of ML in membrane bioreactor (MBR) systems has grown substantially in recent years (Muniz de Queiroz et al. 2025). In particular, previous studies have shown that data-driven models can successfully capture the non-linear dynamics and inhibition effects in anaerobic membrane systems, where process behaviour is strongly influenced by microbial adaptation and toxicity thresholds (Lai et al. 2024; Xu et al. 2026; Yu et al. 2026). As an example, Zhong et al. (Zhong et al. 2022) used a linear model, decision tree, ensemble learning, nearest neighbor model, and support vector machine to predict the water quality of a MBR that treats high-salinity ammonia-nitrogen wastewater. Yaqub and Lee (Yaqub and Lee 2022) used an ensemble model referred to as extreme gradient boosting (XGB) to predict total nitrogen (TN), total phosphorus (TP) and ammonium (NH_4^+) removal from an anaerobic-anoxic-oxic MBR at a wastewater treatment plant, alongside ANNs. To predict effluent (NH_4^+) and TN concentrations in a MBR treating domestic wastewater, Zhuang et al. (Zhuang et al. 2021) evaluated multiple models including regression tree, random forest, adaptive boosting, gradient boosting regression, and deep neural networks. Despite these advances, most existing studies remain focused on predictive accuracy, with limited attention to extracting engineering-relevant insights

such as operational thresholds and system stability limits. Therefore, the application of ML in this study is specifically justified to resolve non-linear inhibition behaviour and to identify the operational window governing COD removal efficiency under phenolic shock conditions. This approach enables the translation of complex process data into actionable engineering insights, which cannot be obtained using conventional modelling approaches.

While ML is widely used in MBRs, a systematic comparison of advanced non-linear techniques as well as interpretability analyses remains limited, particularly when operating Submerged Anaerobic MBRs (SAnMBRs) under high phenolic shock loads for long periods of time. Studies often focus only on maximizing correlation coefficients, failing to investigate why the model works. Accordingly, the objective of this study is not limited to comparing model performance, but to link predictive capability with process understanding by identifying the key drivers and critical thresholds governing reactor stability. In this study, multiple linear regression (MLR), artificial neural networks (ANN), and support vector regression (SVR) are evaluated to predict COD removal efficiency in the treatment of 2,4-dichlorophenol wastewater using a pilot-scale SAnMBR, addressing a key gap in the existing literature. The selection of 2,4-DCP as the target compound is specifically motivated by its well-documented inhibitory effects on anaerobic biological processes and its persistence in industrial effluents, making it a representative model pollutant for assessing system behaviour under phenolic shock loading conditions. Unlike readily biodegradable organics, 2,4-DCP introduces toxicity-driven non-linear responses that manifest at the system level, influencing overall reactor performance and stability rather than directly measured microbial activity (Li et al. 2024). Given the highly non-linear and coupled

Fig. 1 Schematic diagram of the submerged anaerobic membrane bioreactor



nature of the SAnMBR system, particularly under phenolic shock conditions, model interpretability cannot rely on simple coefficient-based or linear sensitivity approaches. Therefore, Partial Dependence Plots (PDPs) are specifically employed to isolate and visualize the marginal effect of individual input variables (e.g., pH and 2,4-DCP) on COD removal while averaging out the influence of other interacting parameters. In parallel, sensitivity analysis is applied to quantitatively rank the relative importance of each variable across its full operational range, enabling identification of dominant drivers under varying conditions. The most effective model is quantified by PDPs and Sensitivity Analysis on the best-performing parameter (e.g., pH, 2,4-DCP concentration). The combined use of these two complementary methods is essential, as PDPs enable the interpretation of non-linear response behaviour and threshold effects, while sensitivity analysis provides a quantitative measure of parameter importance across the operational range, supporting engineering decision-making in complex biological systems (Lai et al. 2024). These interpretability tools enable the translation of model outputs into engineering insights, such as defining safe loading limits, identifying inhibition zones, and distinguishing between biologically controlled processes and fouling-driven operational constraints. To the best of our knowledge, this is the first study to compare these models for long-term SAnMBR treatment of phenolic wastewater. Therefore, this work advances beyond conventional ML applications by integrating prediction with operational decision-making in SAnMBR systems. By combining high-accuracy prediction with deep interpretability, the resulting framework can support the development of real-time control strategies (optimal pH ranges, safe 2,4-DCP loading limits), enhancing process stability, reducing operational expenditure, and advancing the practical digital deployment of SAnMBR technology.

2 Materials and Methods

2.1 Experimental Setup

The experimental setup of the SAnMBR is shown in Fig. 1. The reactor was made of Plexiglas with an effective volume of 8 L and a total volume of 8.733 L. The system was operated as a continuous-flow SAnMBR under long-term dynamic loading conditions (189 days), incorporating a full membrane module and process control strategy representative of practical operation. This SAnMBR was operated over a period of 189 days. The injection of 2,4-dichlorophenol (2,4-DCP) commenced on day 118, after achieving stable baseline reactor performance, in order to systematically evaluate the system response to controlled phenolic shock

loading. The organic loading rate (OLR) was gradually varied between 0.125 and 0.798 kg COD/m³·day to assess system performance under diverse operational conditions. Water-jacket compartments were installed on both sides of the reactor and the reactor temperature was maintained at 36±1 °C by a circulating water bath. This controlled mesophilic temperature was selected to maintain stable anaerobic conditions and eliminate external variability associated with ambient temperature fluctuations. The membrane module was composed of a submerged hollow fiber polypropylene membrane (Hydrol, UK) with an area of 0.1 m². Influent and effluent flows were regulated using two parallel peristaltic pumps (Shiva Amvaj, Iran), controlled via a digital multi-timer (Shiva Amvaj, Isfahan, Iran). To minimize evaporation and salt accumulation in the water bath, distilled water was used, and the bath surface was covered with plastic film.

2.2 Experimental Procedure

The reactor was seeded with 2000 mL of digested sludge obtained from the South Tehran municipal wastewater treatment plant. Prior to seeding, the sludge was thoroughly washed with tap water to remove impurities and characterized for total suspended solids (TSS: 21,082 mg/L) and volatile suspended solids (VSS: 12,937 mg/L).

Synthetic wastewater was freshly prepared daily, containing glucose as the primary carbon source along with essential macro- and micronutrients sourced from Merck (KGaA, Germany). Nutrient concentrations were dosed per gram of COD as follows: NH₄Cl (76.45 mg/g COD), KH₂PO₄ (10.00 mg/g COD), and trace elements including K₂HPO₄ (25.32), FeCl₃ (1.021), CaCl₂·2 H₂O (2.06), MgSO₄·7 H₂O (2.14), MnSO₄·H₂O (0.355), CoCl₂·6 H₂O (0.092), NiSO₄·6 H₂O (0.0763), ZnSO₄ (0.1055), (NH₄) Mo₇O₂₄·4 H₂O (0.4198), CuSO₄·5 H₂O (0.0321), and H₃BO₃ (0.020). The reactor was operated continuously, with 2,4-DCP concentrations gradually increased from 5 mg/L to 300 mg/L following its initial injection. System performance was assessed by monitoring COD and 2,4-DCP removal efficiencies under different loading regimes.

2.3 Analytical Methods

Physicochemical parameters including temperature, pH, COD, turbidity, TSS, TDS, VSS, 2,4-DCP concentration and electrical conductivity (EC) were monitored throughout the study. COD and phenol concentrations were determined using the closed reflux colorimetric method (APHA 5220D) and direct photometric method (APHA 5530D), respectively, with a DR 6000 spectrophotometer (Hach, Germany). Turbidity was measured using the nephelometric method with a Hach 2100AN turbidimeter..

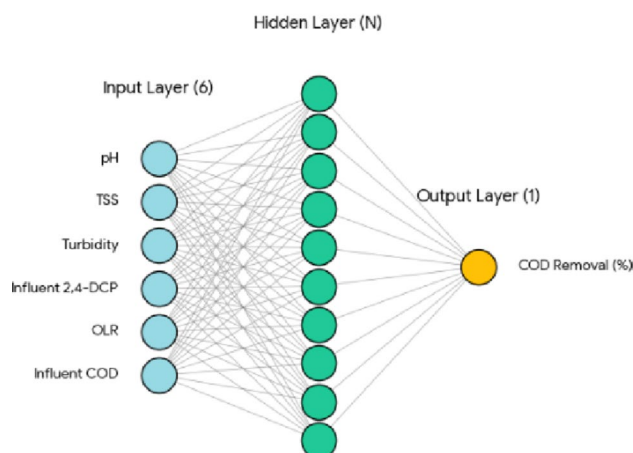


Fig. 2 Architecture of the ANN model (6-N-1 structure)

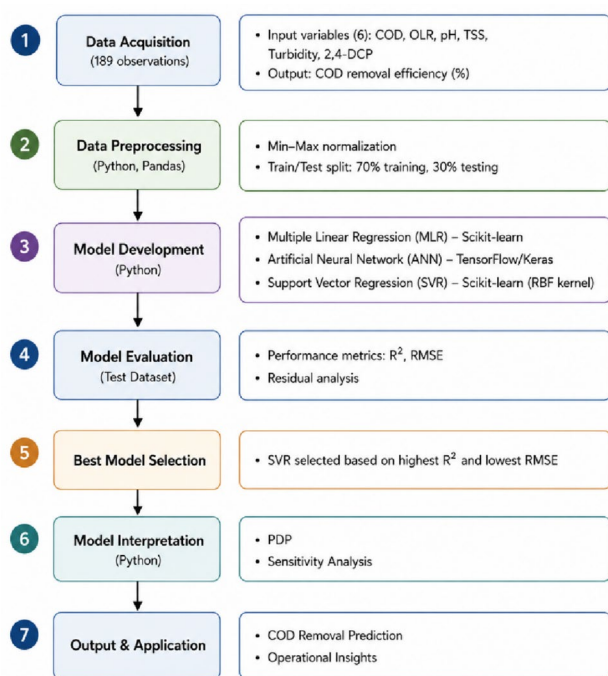


Fig. 3 Schematic workflow of the machine learning modelling process, including data preprocessing, model development, evaluation, and interpretability analysis

pH was adjusted using 0.1 N H_2SO_4 and 0.1 N NaOH and measured with a multi-parameter analyser (CONSORT C831, Belgium). TSS, TDS, and VSS were analysed in accordance with APHA Standard Methods 2540 C, 2540D, and 2540E. For TSS, 25 mL samples were filtered through pre-weighed Whatman GF/C filters (UK), dried at 103–105 °C for 24 h (Mettler oven), cooled in a desiccator, and reweighed. TDS was determined by evaporating 25 mL samples at 180 °C until a constant weight was achieved.

VSS was calculated by igniting the TSS filters at 550 °C for 30 min in a muffle furnace (Nabertherm, Germany), followed by cooling and reweighing. All measurements were conducted in triplicate, and average values were reported.

2.4 Data Acquisition and Pre-processing

The predictive modelling analysis was based on an experimental dataset collected throughout the operational period of the Submerged Anaerobic Membrane Bioreactor (SAnMBR). In total, 189 valid data points were obtained, with no missing entries. The selection of input features (X) was guided by their known relevance to biological wastewater treatment and their operational importance in the SAnMBR process. Six independent variables were used in the modelling: influent COD (mg/L), organic loading rate (OLR, $\text{gCOD}/\text{m}^3\cdot\text{day}$), influent 2,4-dichlorophenol (mg/L), influent turbidity (NTU), influent TSS (g/L), and influent pH. The dependent variable (Y) was the COD removal percentage (%). To ensure reliable model training and evaluation, the dataset was randomly divided into two parts: 70% (around 132 samples) for model training and the remaining 30% (around 56 samples) for testing. This division allowed the trained models to be evaluated on data they had not previously encountered, providing a fair assessment of their generalization capability. Prior to normalization, the dataset was first split into training and testing subsets. The Min–Max scaling parameters were then fitted using only the training data and subsequently applied to both training and testing sets. All input features were normalized using the Min–Max scaling method to rescale their values to the [0,1] range. This normalization step avoided bias toward variables with larger numerical ranges and ensured that each input contributed equally during training.

2.5 Predictive Machine Learning Model Implementation

Three distinct machine learning algorithms, Support Vector Regression (SVR), Artificial Neural Networks (ANN), and Multiple Linear Regression (MLR), were implemented and benchmarked to predict the COD removal efficiency.

2.5.1 Multiple Linear Regression (MLR)

Regression models are widely used in a variety of fields to define relationships among multiple parameters; resulting equations can then be applied to water and wastewater treatment problems. In order to evaluate the inherent linearity of the SAnMBR system's response, the MLR model served as the foundational baseline. MLR uses an ordinary least-squares method in order to establish a simple linear

relationship between the six input variables (X) and the target output (Y). The MLR model is represented by the linear Eq. (1) (Gyparakis et al. 2025):

$$Y = \beta_0 + \sum_{i=1}^6 \beta_i x_i \quad (1)$$

Here, Y represents the predicted COD removal percentage, x_i are the independent parameters, β_i are the corresponding constant coefficients (or weights), and β_0 is the intercept. The primary purpose of MLR was to provide a quantitative benchmark against which the performance gains achieved by the non-linear SVR and ANN models could be judged.

2.5.2 Support Vector Regression (SVR)

In order to address the complex, non-linear nature of environmental datasets, SVR was selected as the primary non-linear predictive tool. This model was chosen because of its high generalization capacity, which is crucial when using complex, non-linear environmental datasets. This model uses a kernel function to map the six input features to a high-dimensional space, converting the non-linear regression problem into a linear one. As a result of rigorous optimization, the Radial Basis Function (RBF) kernel was selected, as it was most effective at capturing biological treatment dynamics. The model's performance was fine-tuned by optimizing the key hyperparameters, namely the penalty parameter (C) and the kernel width (γ) (Najafzadeh and Niazmardi 2021).

2.5.3 Artificial Neural Network (ANN)

Artificial neural network concepts date back to early artificial neuron models introduced in the 1940s; is fundamentally based on simulating the structure and function of biological nervous systems. As a result of the introduction of the

backpropagation training algorithm, artificial neural networks can represent complex, non-linear functionality using fewer parameters. This capability has led to extensive use in a variety of fields, including water quality modelling, medicine, and engineering (Reza et al. 2024). In order to balance the complexity against the risk of overfitting, the developed MLP network structure (6-N-1) was systematically tuned to determine the optimal number of neurons (N) in the hidden layer. The network architecture, which defines the flow of information through the system, is visually represented in Fig. 2. The architecture consists of: (1) an Input Layer with 6 neurons, receiving the normalized input features; (2) one Hidden Layer using the *tansig* (tangent sigmoid) transfer function to introduce non-linearity; and (3) a single Output Layer neuron using the *purelin* (linear) function to yield the final COD Removal prediction. The network was trained in a standard computational environment (Python), replicating the desired output based on unseen inputs in accordance with the provided training data.

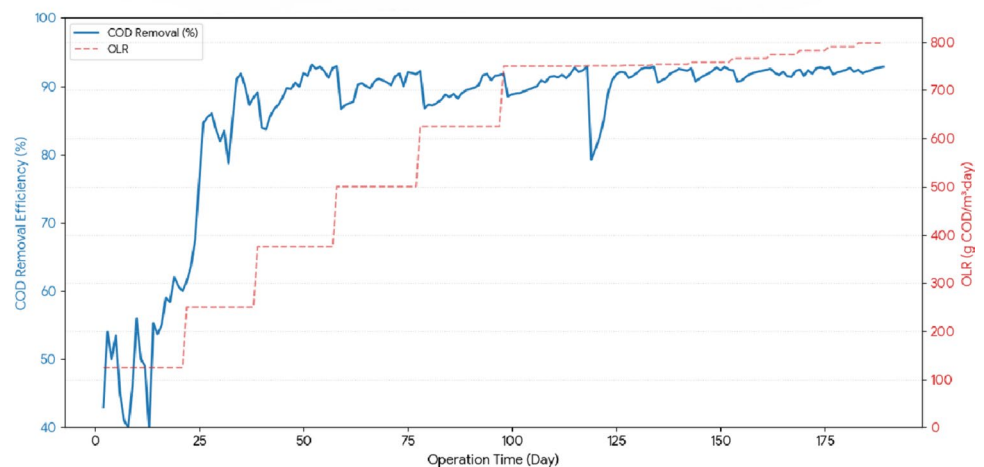
2.5.4 Computational Tools and Environment

All machine learning models, including SVR, ANN, and MLR, as well as the interpretability analyses such as Partial Dependence Plots (PDP) and sensitivity analysis, were implemented using Python (Version 3.10). The modelling tasks were performed using the Scikit-learn library (version 1.2.2) for SVR and MLR, while the ANN model was developed using the TensorFlow/Keras framework. Data preprocessing, including normalization and dataset splitting, was carried out using the Pandas library, and visualization was performed using Matplotlib and Seaborn.

The overall modelling workflow is illustrated in Fig. 3, which summarizes the sequential steps from data acquisition to model interpretation.

As shown in Fig. 3, the modelling workflow follows a sequential process from data acquisition to model

Fig. 4 Time series of COD removal efficiency and OLR in the SAnMBR system



interpretation. The process begins with experimental data collection (189 observations), followed by data preprocessing, including Min–Max normalization and division of the dataset into training (70%) and testing (30%) subsets. Subsequently, three machine learning models (MLR, ANN, and SVR) were developed to capture the relationship between input variables and COD removal efficiency. Model performance was then evaluated using statistical metrics, including the coefficient of determination (R^2) and root mean square error (RMSE), along with residual analysis. Based on these criteria, the best-performing model (SVR) was selected for further analysis.

The modelling process consisted of data preparation, model training using the training dataset, and performance evaluation using an independent test dataset based on R^2 and RMSE metrics. For model interpretation, PDPs were generated to examine the influence of individual input variables on the predicted COD removal by varying one parameter while keeping others unchanged within the dataset context. In addition, sensitivity analysis was conducted by systematically varying each input variable within its observed range and evaluating the corresponding changes in the model output to quantify the relative importance of the parameters.

2.6 Model Evaluation Criteria

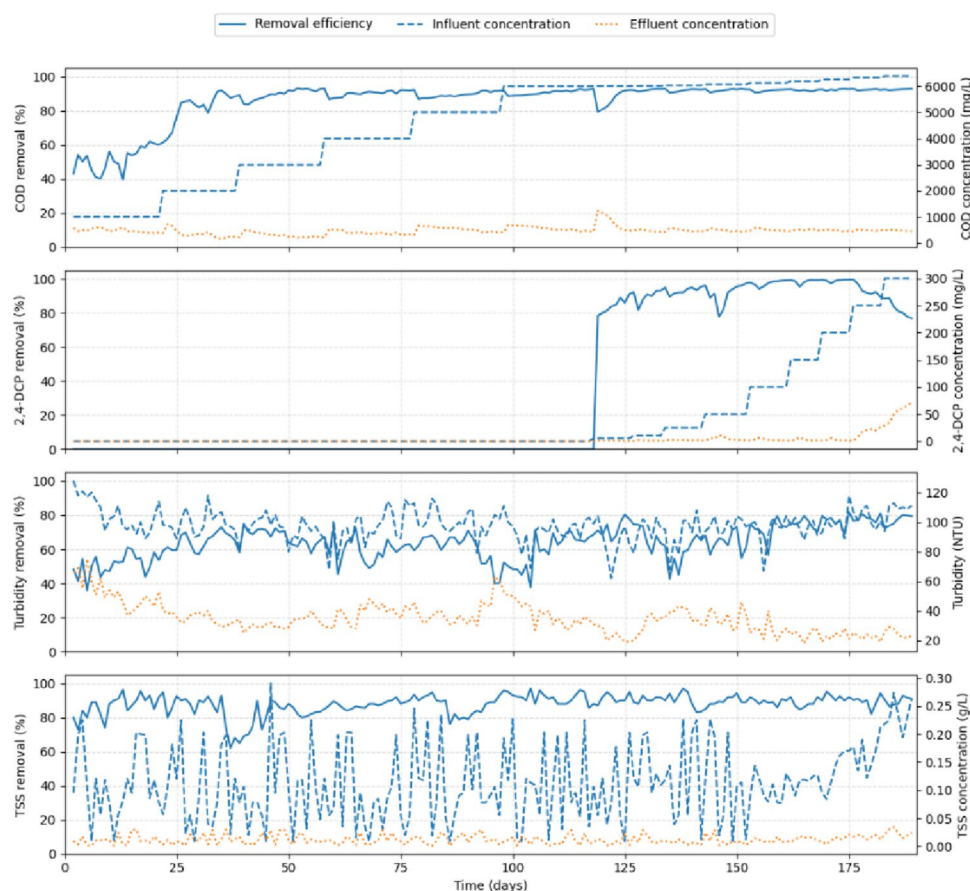
In order to assess the predictive accuracy and robustness of the MLR, ANN, and SVR models, an independent test set was utilized. The coefficient of determination (R^2) and the root mean square error (RMSE) were used as two universally accepted statistical metrics. The R^2 value is defined by Eq. (2):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

where y_i represents the observed values, $\hat{y}_i$ denotes the predicted values, $\bar{y}$ is the mean of the observed values, and n is the total number of data points. A positive R^2 value indicates that a portion of the target variable's variance is successfully predicted by the model, indicating a good fit (values close to 1 are considered ideal). The RMSE is calculated as Eq. (3):

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3)$$

Fig. 5 Temporal variation of removal efficiencies (left axis) and corresponding influent and effluent concentrations (right axis) of COD, 2,4-dichlorophenol (2,4-DCP), turbidity, and TSS during the 189-day operation of the SANMBR system.



where the terms have the same meaning as defined above. RMSE quantified the average magnitude of prediction errors, with lower values indicating higher predictive accuracy.

3 Results and Discussion

3.1 SAnMBR Experimental Performance

The 189-day time series (Fig. 4) shows that COD removal rose during an initial acclimation period (Days 1–50) from ~43% to roughly 85–90% and then remained mostly between 80% and 95% despite large, staged increases in OLR (from ~125 to ~800 g COD m⁻³·d⁻¹). Several non-exclusive mechanisms can explain this behaviour.

A core advantage lies in membrane retention, which effectively decouples the Solids Retention Time (SRT) from hydraulic retention time (HRT). This separation enables the accumulation of slow-growing biomass, particularly methanogens, allowing the system to sustain efficient conversion of soluble organic matter (COD) into methane. As a result, stable COD removal is maintained under varying loading conditions, and the impact of transient hydraulic and organic stress is significantly reduced. This critical SRT effect has been repeatedly observed as a primary advantage of SAnMBR configurations (Robles et al. 2018). Furthermore, the maintained mesophilic temperature (36 ± 1 °C) and high mixed liquor concentrations typical of long-term MBR systems enhance COD degradation kinetics by stabilizing microbial activity and promoting enzyme-mediated reactions involved in hydrolysis and methanogenesis. Under mesophilic conditions, substrate utilization rates are more consistent and less sensitive to fluctuations, which supports

efficient conversion of soluble organic matter and reduces the risk of kinetic limitations at elevated OLRs (Khan et al. 2024). As a result, higher apparent degradation rates and improved process stability are achieved under variable loading conditions. System-level control strategies are also key contributors to resilience. This includes controlled feed step-changes and proactive fouling management (such as gas sparging, regular backwash, and chemical cleaning), all of which limit the likelihood that transient high loads or membrane fouling translate into long-term performance loss.

Additionally, the observed stability is partially attributable to microbial community adaptation. Under prolonged exposure to persistent substrates and phenolic compounds, a selective pressure is applied, which can increase the relative abundance of degraders and methanogens adapted to these recalcitrant substrates. Representative genera reported in similar anaerobic systems include syntrophic bacteria such as *Syntrophus* and *Syntrophorhabdus*, which are involved in the initial breakdown of aromatic compounds, and methanogenic archaea such as *Methanosaeta* and *Methanosarcina*, responsible for the final conversion of intermediates to methane (Muñoz Sierra et al. 2020). These microbial functions are consistent with those reported in comparable phenolic wastewater treatment systems. Where such favourable adaptation occurs, reactors show improved stability, which is highly consistent with the high removal rates seen in this campaign. However, it is important to note that performance can significantly decline past reactor-specific OLR thresholds.

Figure 5 illustrates the variation in COD, 2,4-DCP, turbidity, and TSS removal efficiencies during stepwise increases in the organic loading rate (OLR).

The COD removal rate remained high throughout the experiment (85–95%), with short-term decreases of 5–8% following each increase in OLR, followed by rapid recovery. There was a transient accumulation of soluble microbial products (SMP) and extracellular polymeric substances (EPS) during stress conditions, which can cause these temporary drops. The EPS produced and released by microorganisms, together with SMP, can accumulate on membrane surfaces, causing fouling and limiting substrate diffusion. Comparable EPS and SMP accumulation has also been observed in AnMBRs operated under fluctuating organic loads and temperatures, where it was linked to reduced biodegradation performance (AbuKhadra et al. 2024; Medina et al. 2020).

Turbidity removal improved progressively from 60–65% in the early phases to approximately 75–80% in later stages, reflecting the gradual maturation of biomass and enhanced floc formation that improves colloid capture. TSS removal remained consistently high (>90–95%) across all phases,

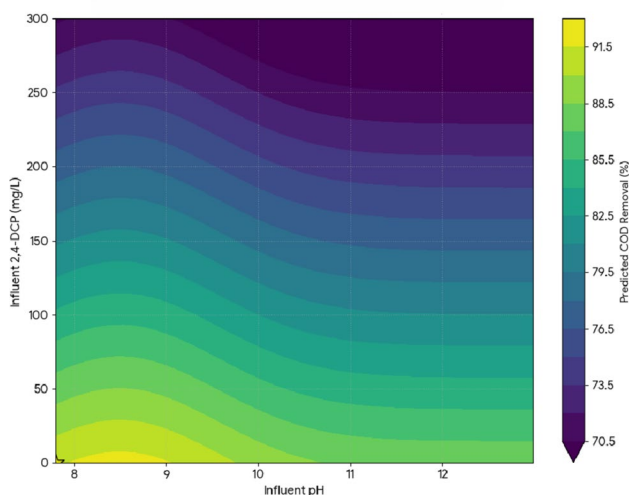


Fig. 6 Two-way partial dependence plot: influent pH versus influent 2,4-DCP

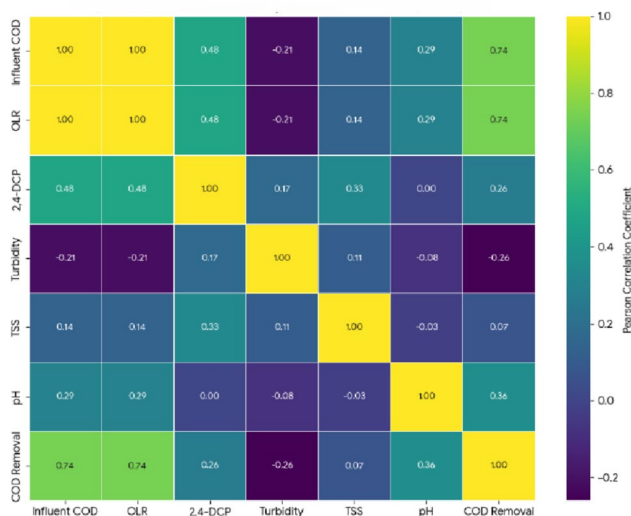


Fig. 7 Pearson correlation heatmap showing the relationships between the six input variables and COD removal efficiency in the SANMBR system.

Table 1 Comparative performance of the ANN, SVR, and MLR models based on statistical indicators (R^2 and RMSE) for the training and testing stages

Model	R^2 (Train)	R^2 (Test)	RMSE (Train)	RMSE (Test)
ANN	0.603	0.397	7.89	10.56
SVR	0.975	0.952	1.97	2.96
MLR	0.734	0.717	6.45	7.23

demonstrating the membrane's reliable capacity for solids retention even during load disturbances. This stability aligns with findings from recent SANMBR and AnMBR research, where membrane separation effectively maintains effluent clarity and solids polishing despite fluctuations in organic removal efficiency (He et al. 2020; Pacheco-Ruiz et al. 2021).

The removal of 2,4-DCP exhibited a more dynamic trend than bulk COD. During the initial period, removal efficiency was relatively low (<20–30%), but after microbial acclimation it increased steadily and stabilized at 80–95%. Minor reductions were observed after OLR increases, which were followed by rapid recovery. This lag compared to COD removal is typical for chlorinated phenolic compounds, where dechlorination and aromatic ring-cleavage steps are kinetically limiting and highly sensitive to substrate or toxicity shocks until dehalogenating microbial populations become fully established (Geng et al. 2020). Under longer hydraulic retention times (HRTs) and steady operating conditions, AnMBRs have been shown to achieve parallel high removals of COD and chlorophenols, as shown by Zhu et al. (2020), who reported average COD and 2-chlorophenol removals of approximately 93% and 94%, respectively, at HRTs of 48–96 h for an influent 2-chlorophenol concentration of 5–200 mg/L (Zhu et al. 2020).

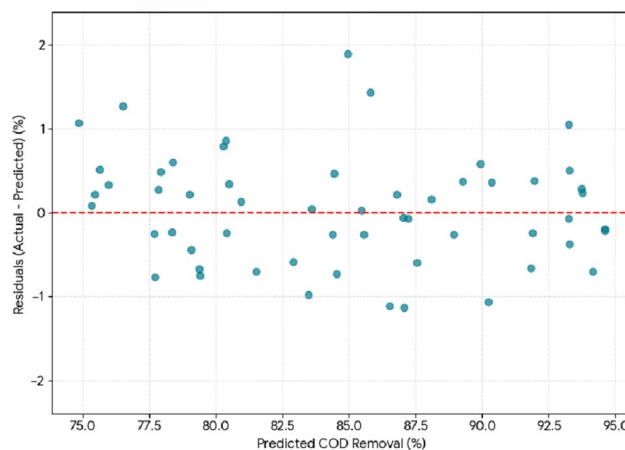


Fig. 8 Residual Analysis of the Optimal SVR Model (Test Set). The plot shows the distribution of prediction errors (residuals) against the SVR model's predicted COD removal values

When compared with recent literature on SANMBR and AnMBR systems, the steady-state removal efficiencies observed in this study fall within the upper range of reported values, and the transient fluctuations are consistent with expected operational dynamics. A 189-day SANMBR experiment treating 2,4-DCP wastewater achieved almost complete removal at stable load (COD≈99.6%, 2,4-DCP≈99.6%), confirming that once microbial communities are acclimated, phenolic degradation can reach efficiencies comparable to bulk organics. Other hybrid and submerged anaerobic membrane configurations have similarly demonstrated strong solids capture and high COD removal, with transient performance variations primarily governed by OLR step changes, HRT, and fouling management strategies (Wang et al. 2015). The slight differences between the present results and the plateau values reported in the literature may reflect differences in OLR ramping strategy, acclimation duration, phenolic compound type (2,4-DCP versus 2-CP), and fouling behaviour driven by SMP and EPS accumulation (Chen et al. 2017).

The two-dimensional partial dependence analysis was performed to investigate the complex bivariate interaction between influent pH and 2,4-DCP concentration, which were identified as the most influential parameters based on the sensitivity analysis results (Fig. 6), where they exhibited the highest relative impact compared to other operational variables. This analysis extends the interpretation beyond single-factor effects by examining how interacting operational variables influence SANMBR performance. As shown in Fig. 6, the highest predicted COD removal efficiency (92–95%) is achieved only under the strict combination of low 2,4-DCP concentrations (<50 mg/L) and the optimal pH range (pH 8–9). This confirms the primary role of pH in maximizing microbial enzyme activity (Zhang et al. 2023).

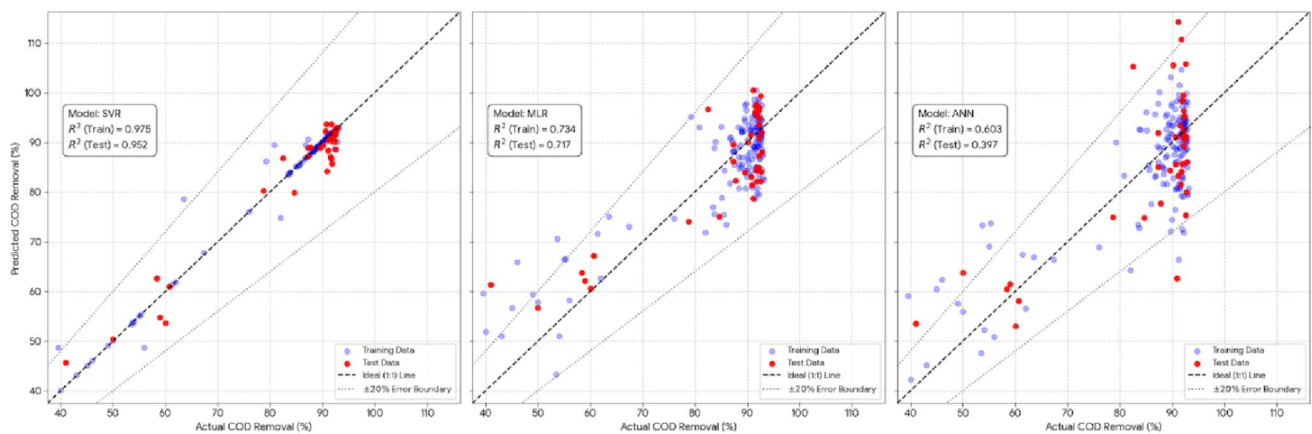


Fig. 9 Comparison of actual (experimental) and predicted COD removal (%) by the SVR, MLR, and ANN models. The R^2 values for the training phase (blue data) and testing phase (red data) are shown

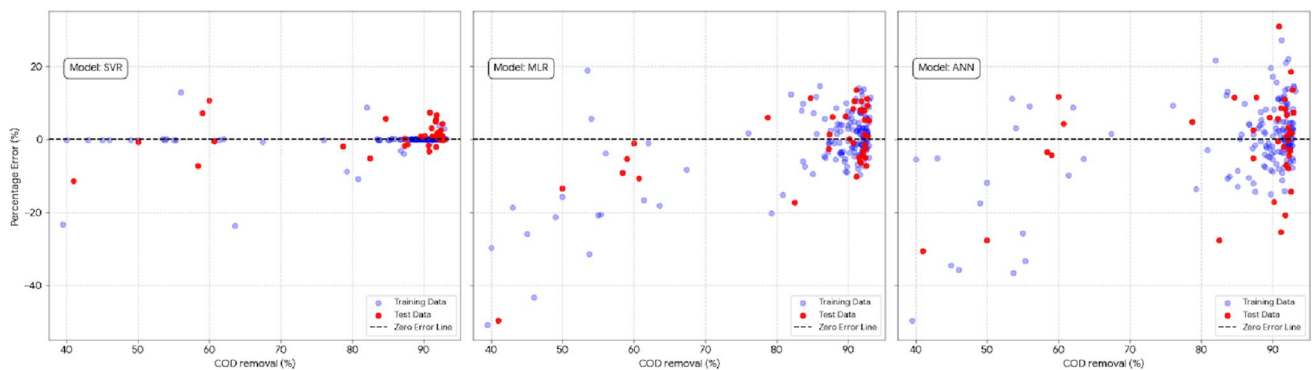


Fig. 10 Distribution of Percentage Error (%) versus Actual COD removal (%) for the SVR (left), MLR (center), and ANN (right) models. Training and testing data are represented by blue and red points, respectively

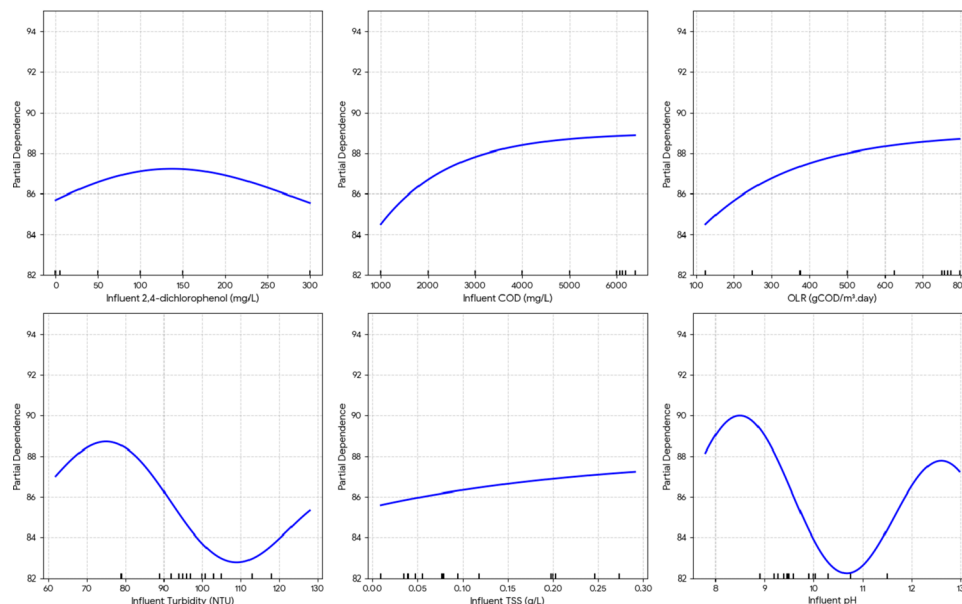
Crucially, the plot reveals a non-linear, compounding inhibitory effect outside this optimal zone. The steepest decline in COD removal is observed along the increasing 2,4-DCP concentration axis. While performance remains relatively high when 2,4-DCP is low (regardless of minor pH changes), performance rapidly deteriorates as 2,4-DCP concentration increases above 150 mg/L. This deterioration is exacerbated outside the optimal pH zone. This behaviour is likely due to two factors: (1) Low pH (acidic conditions) increases the proportion of 2,4-DCP present in its undissociated, non-ionized form, which readily penetrates microbial cell membranes, dramatically increasing its toxicity; and (2) pH values outside the methanogenic ideal already stress the system, making the biomass less resilient to additional chemical inhibition (Fernandez et al. 2022; Zhang et al. 2023). This finding has direct operational significance for the SAnMBR system, as it demonstrates that optimization is not merely about minimizing 2,4-DCP but is a dynamic trade-off. Maintaining rigorous pH control provides a critical buffering mechanism against moderate toxic spikes.

3.2 Correlation Analysis of Model Variables

The correlation matrix (Fig. 7) was instrumental in evaluating the linear dependencies among the six selected input parameters and the target output, COD Removal Percentage, serving as a critical step before non-linear machine learning modelling for the SAnMBR system.

Influent COD and OLR showed a very strong positive correlation ($r = 1.00$), which is expected because OLR is partly derived from influent COD. Although such a relationship may cause multicollinearity in linear models, both variables were retained in the SVR model to preserve the operational description of the SAnMBR system. Regarding the target output, influent COD and OLR showed the strongest positive linear associations with COD removal ($r = 0.74$), while pH showed a weaker but still positive correlation ($r = 0.36$). Conversely, influent TSS showed an almost negligible correlation with COD removal ($r = 0.07$), whereas turbidity exhibited a weak negative association ($r = -0.26$). This suggests that suspended solids alone were not a primary linear driver of COD removal, while increased turbidity may be linked to conditions that reduce treatment

Fig. 11 PDPs for the 6 numerical input parameters based on the SVR



performance. Influent 2,4-DCP showed a weak positive correlation with COD removal ($r = 0.26$), indicating that its direct linear association with treatment performance was limited. Therefore, correlation analysis alone may not adequately describe the role of 2,4-DCP, particularly if its effect depends on concentration thresholds or interactions with other operational variables, as further examined through PDP analysis. Influent COD and OLR showed the strongest positive correlations with COD removal ($r = 0.74$ for both), followed by influent pH with a moderate positive correlation ($r = 0.36$). These results suggest that organic loading variables had the closest linear association with COD removal, while pH showed a weaker but still noticeable relationship.

3.3 Model results for COD Removal Prediction

The predictive performance of the three developed models (SVR, MLR, and ANN) was evaluated using the coefficient of determination (R^2) and Root Mean Square Error (RMSE), as presented in Table 1.

The SVR model demonstrated the most robust and accurate predictions, achieving the best results for the testing dataset ($R^2 = 0.952$ and $RMSE = 2.96$). Whereas for the training data, as demonstrated by Table 1, SVR also shows the best results for the chosen statistical indicators ($R^2 = 0.975$ and $RMSE = 1.97$). The significant difference between the ANN model's performance on the training data ($R^2 = 0.603$) relative to its testing data ($R^2 = 0.397$) is likely due to the limited predictive capacity of the. The effects of this become apparent when observing the model's performance in the testing dataset, as its performance was exceeded by both the MLR and SVR models.

To further validate the reliability and non-systematic nature of the errors produced by the optimal SVR model, a residual analysis was performed. The residual plot (Fig. 8) displays the difference between the actual and predicted COD removal values (the residual error) plotted against the model's predicted output for the independent test set.

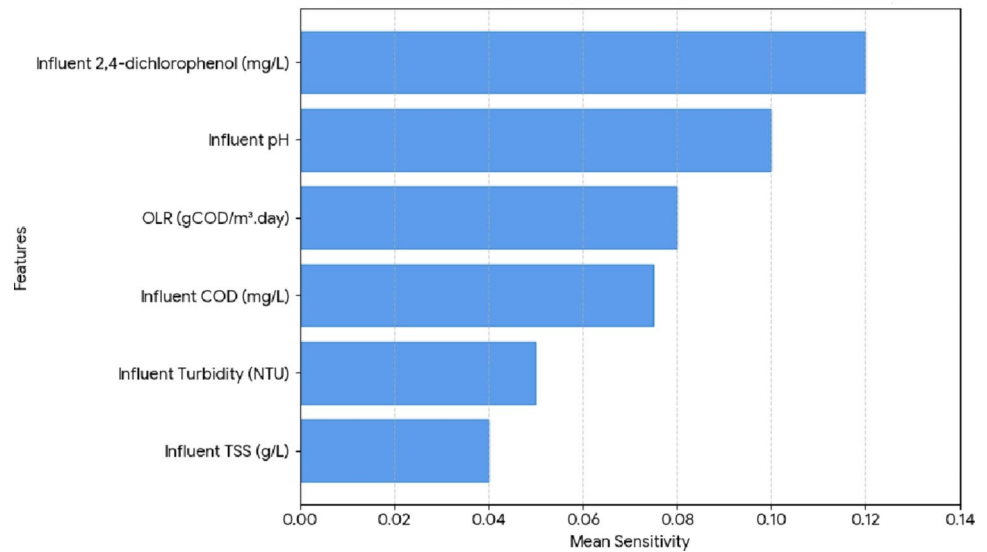
The distribution of data points exhibits a random, uniform scatter around the horizontal zero line ($y = 0$). The absence of any discernible patterns such as a cone shape, a trend, or clustering at specific predicted values is a critical finding. This confirms that the SVR model's high predictive accuracy ($R^2 = 0.952$) is robust and that the model has successfully accounted for all systematic dependencies within the SAnMBR process, suggesting that the remaining errors were largely random.

Furthermore, Fig. 9 shows the residual plots with percentage errors for the models. This plot validates the SVR's accuracy, most of its percentage errors (both blue and red points) are distributed randomly and very close to the zero-error line.

The ANN and MLR models, however, show a much wider distribution of errors (Fig. 10), with the ANN model exhibiting clear non-random patterns (heteroscedasticity), reinforcing its unsuitability.

The test-set R^2 for the SVR model based on the testing dataset was 0.952 (or 95.2%), clearly demonstrating its superior ability to model the non-linear SAnMBR process. The training and testing data points are represented using blue and red, respectively. The plots also have a visual aid in the form of a 20% error bound (gray dotted lines) to indicate relative deviation between actual and predicted results. Here, the SVR model is shown to excel in its prediction

Fig. 12 Sensitivity analysis plot for six input parameters used in the SVR model



ability. In its corresponding parity plot in Fig. 9, the testing data points (red) are tightly clustered around the ideal 1:1 line and fall almost entirely within the error bounds. In contrast, the ANN model's parity plot shows significant scatter and multiple data points outside this bound, confirming the overfitting observed in Table 1.

3.3.1 Partial Dependence Plots (PDPs)

PDPs are used to visualize the interaction between individual input features and their respective effect on the machine learning model's output prediction, assuming all other features remain constant. These plots illustrate the impact of changing a single feature on the prediction of the dependent variable, thereby making the results of the advanced SVR model more interpretable and understandable. Fig. 11 shows the PDPs for six key numerical input parameters employed in the current study.

The PDP for influent 2,4-dichlorophenol (2,4-DCP) concentration reveals a non-linear response pattern. An initial increase in partial dependence values is observed within the lower concentration range (0–150 mg/L), indicating that the model predicts relatively stable or slightly improved COD removal performance in this interval. This behaviour suggests that, within this range, the system is not immediately limited by toxicity effects. However, beyond 150 mg/L, the trend gradually plateaus and begins to decline, reflecting the onset of inhibitory effects on microbial activity. This transition highlights the balance between substrate availability and toxicity, where increasing pollutant concentration initially does not suppress system performance but eventually leads to inhibition, particularly affecting methanogenic processes (Li et al. 2024).

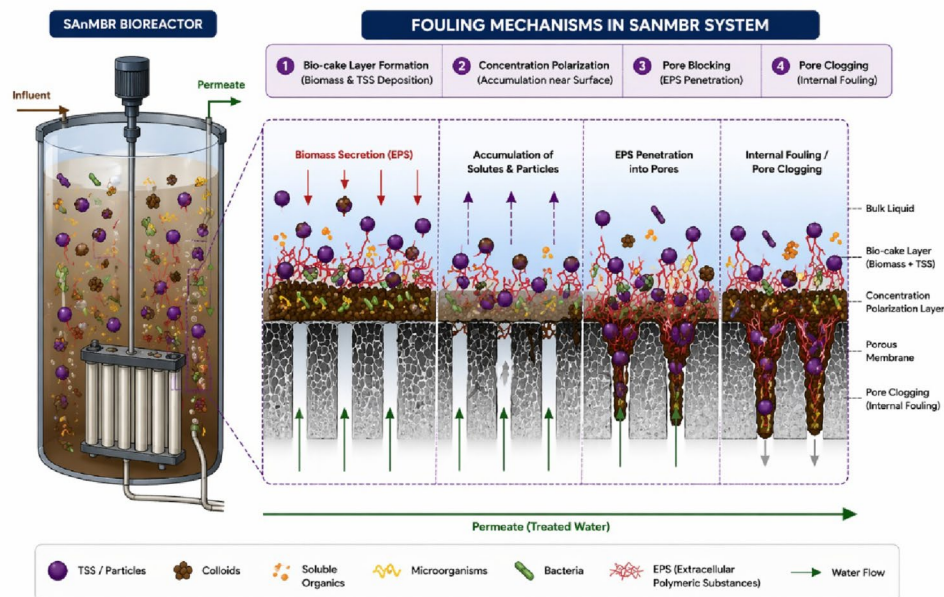
For influent pH, the PDP showed a clear non-linear relationship, with the highest predicted COD removal efficiency observed in the pH range of approximately 8–9. Outside this optimal window, particularly under acidic (pH < 7) or strongly alkaline (pH > 9) conditions, the predicted performance declines. This trend is consistent with the known sensitivity of anaerobic microbial communities to pH, where deviations from the optimal range negatively affect enzymatic activity and metabolic stability (Meegoda et al. 2018).

Comparatively, influent COD (mg/L) and OLR (gCOD/m³.day) exhibit relatively minor influence on the predicted COD removal efficiency within the studied ranges, indicating that the system operated below its critical loading threshold. Similarly, influent turbidity (NTU) and TSS (g/L) show a slight negative trend, where increasing values correspond to marginal decreases in predicted efficiency. This is consistent with the rate-limiting nature of particulate hydrolysis in anaerobic systems, as particulate matter requires additional processing steps compared to soluble substrates (Wang et al. 2022).

3.3.2 Sensitivity Analysis

This section quantifies the influence of individual input features on the optimal SVR model's prediction of COD removal using univariate sensitivity analysis framework. Unlike model-specific explainability methods such as SHapley Additive exPlanations (SHAP), which rely on internal model structures or game-theoretic formulations, the present approach is a model-agnostic sensitivity analysis designed to evaluate the response of the trained model to controlled variations in input variables. This approach provides a direct and computationally efficient assessment of global feature influence, suitable for identifying dominant

Fig. 13 Schematic illustration of membrane fouling mechanisms in the SANMBR



process variables. In this approach, each input feature was varied independently across its observed range using a systematic discretization procedure, while all other input variables were held constant at their baseline values. Specifically, the baseline inputs were defined using the mean values of the original dataset, and each variable was perturbed across its minimum–maximum range using a fixed number of uniformly spaced intervals. For each variable, a series of uniformly spaced points within its range was generated, and the trained SVR model was used to predict COD removal for each modified input set. The sensitivity index was then calculated as the mean absolute deviation between the predicted outputs obtained from the perturbed inputs and the baseline predictions. This metric quantifies the extent to which variations in each input feature influence the model output. Higher sensitivity values indicate stronger influence, while values approaching zero suggest minimal impact. The sensitivity analysis plot (Fig. 12) shows the relative importance of each input variable, with 2,4-DCP concentration and pH emerging as the main factors influencing predicted COD removal.

The analysis clearly establishes the crucial role of specific input parameters. Influent 2,4-dichlorophenol (mg/L) exhibits the highest sensitivity, confirming its status as the most influential feature in the SVR model. This finding is highly consistent with the partial dependence plot (PDP) results, which showed a strong negative impact on COD removal (a predicted decline from a stable 90% to below 70%) as its concentration increased. The high sensitivity to 2,4-DCP highlights the need to control inhibitory compounds, as this response is consistent with the known toxicity of chlorophenols toward methanogenic archaea in anaerobic systems.

Influent pH is the second most influential parameter, which accurately reflects the SANMBR system's biological nature and its dependence on a narrow optimal range. The PDPs demonstrate that removal efficiency is maximized within an optimal pH range of approximately 8–9, beyond which system performance progressively declines due to reduced microbial activity under non-optimal conditions. Furthermore, parameters related to bulk organic load, such as influent COD and OLR, displayed lower sensitivity. Their limited influence is consistent with the PDPs showing largely flat impact curves, suggesting that the SANMBR system was highly robust and operated below its peak loading capacity throughout the dataset range. Influent turbidity and influent TSS had the lowest sensitivity scores, indicating that changes in these variables had only a minor effect on the SVR-predicted COD removal. This may be because particulate matter affects anaerobic treatment mainly through hydrolysis, which is typically slower than the degradation of soluble organic compounds. As a result, marginal changes in TSS or turbidity were less influential than soluble inhibitory factors such as 2,4-DCP.

3.4 Operational Implications and Fouling Risk Analysis

The interpretation of the SVR model provides insight into both biological treatment performance and membrane-related operation within the SANMBR system. Although influent TSS and turbidity showed low sensitivity toward predicted COD removal, their role should be interpreted within the broader operating context rather than dismissed as irrelevant. Within the investigated range, the low sensitivity

of TSS and turbidity indicates that particulate-related variables had a limited effect on SVR-predicted COD removal compared with pH and 2,4-DCP concentration. This observation is specific to the operating conditions investigated in this study and should not be generalized across anaerobic treatment systems. COD comprises both soluble and particulate fractions, and the hydrolysis of particulate organic matter is often the rate-limiting step in anaerobic digestion. In SAnMBRs, however, extended solids retention and effective biomass retention can promote particulate hydrolysis, which may explain why TSS and turbidity showed limited influence on predicted COD removal under stable operating conditions (Sandoval-García et al. 2025; Turan 2025).

Despite their relatively minor contribution to predicted COD removal, particulate-related parameters remain critically important from an operational perspective, particularly in relation to membrane fouling behaviour. As illustrated in Fig. 13, membrane fouling in the SAnMBR system is governed by multiple interacting mechanisms rather than a single dominant process. Initially, the accumulation of suspended solids and colloidal matter near the membrane surface leads to concentration polarization, which increases local solute concentration and promotes further deposition. This is followed by the formation of an external bio-cake layer composed of biomass, TSS, and extracellular polymeric substances (EPS). Over time, EPS production and microbial activity facilitate the penetration of organic matter into membrane pores, resulting in pore blocking and, ultimately, internal pore clogging. These sequential and overlapping mechanisms contribute to increased hydraulic resistance and progressive decline in membrane performance, leading to elevated transmembrane pressure and more frequent maintenance requirements (Du et al. 2020; Zhang and Hao 2022).

These results highlight a fundamental decoupling between biological treatment efficiency and operational sustainability. While the SVR model identifies COD removal as primarily governed by biochemical and inhibitory factors (e.g., pH and 2,4-DCP concentration), particulate matter exerts a dominant influence on membrane fouling dynamics and long-term system stability. The low sensitivity of TSS in the SVR model should therefore not be interpreted as operational insignificance, but rather as an indication that fouling-related impacts are not captured by COD-based performance metrics.

From an operational standpoint, this dual behaviour underscores the necessity of adopting a multi-objective optimization strategy. Optimization based solely on COD removal may lead to suboptimal operating conditions, where membrane fouling accelerates despite stable treatment performance. Although increasing TSS may not significantly affect COD removal efficiency within the studied range, it

directly contributes to more frequent cleaning, increased chemical consumption, and elevated operational costs (Ma et al. 2015). This suggests that, beyond a critical particulate threshold, the system transitions from a performance-controlled regime to a fouling-controlled regime, even when apparent treatment efficiency remains unchanged. Therefore, controlling particulate load is essential not only for fouling mitigation but also for maintaining overall system efficiency and cost-effectiveness. These findings emphasize that optimal SAnMBR operation cannot be defined solely by treatment performance indicators, but must incorporate fouling propensity, membrane resistance development, and operational resilience under variable process conditions.

4 Conclusion

This study shows that a data-driven modelling approach can capture the non-linear behaviour of SAnMBR systems under phenolic shock conditions, with SVR providing reliable prediction of COD removal in a biologically sensitive process. Model interpretation showed that performance was strongly affected by the interaction between pH and inhibitory phenolic compounds, suggesting that stable operation depends not only on organic loading but also on keeping toxicity below critical thresholds. The decline in biological performance at elevated 2,4-DCP concentrations further indicates that high-strength phenolic wastewater may require dilution, pretreatment, or selective recovery of phenolic compounds before biological treatment. This agrees with previous studies showing that detoxification can improve microbial activity and enhance subsequent anaerobic treatment. The results also show that biological removal and membrane operation are controlled by different factors. COD removal was mainly influenced by biochemical conditions, whereas particulate matter remains important for membrane fouling and long-term process stability. Although phenolic compounds were treated as inhibitory pollutants in this study, their recovery before biological treatment is gaining attention because of their potential value as antioxidants and industrial precursors. These findings support the use of interpretable machine-learning models for digital monitoring and adaptive control of SAnMBR systems treating complex industrial wastewater. Future work should integrate the developed SVR model into real-time control frameworks to support dynamic adjustment of key variables such as pH and influent 2,4-DCP concentration. Including fouling-related indicators, such as EPS, SMP, and transmembrane pressure, would also allow multi-objective optimisation of biological performance and membrane operation.

Author Contributions Milad Mousazadehgavan: Conceptualization, Writing – original draft, Software, Methodology, Investigation, For-

mal analysis, Project administration, Validation. B. Abdullhadi: Writing – original draft, Investigation, Writing – review & editing. Milad Basirifard: Software, Formal analysis, Writing – original draft. Farideh Malekdar: Writing – original draft, Writing – review & editing. Mahsa Shahi Jouneghani: Writing – original draft, Investigation, Formal analysis. Adel Kamyab Rudsari: Conceptualization, Data Collection, Writing – original draft, Methodology, Investigation. Reza Ghanbari: Writing – review & editing, Supervision.

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Data Availability The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing Interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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